

4 Rice, *Oryza sativa*, is an important food crop for human consumption.

Two varieties of rice are T2A-1 and Shanyou 2.

Shanyou 2 has been developed in the traditional way by selective breeding (artificial selection) and T2A-1 is an example of a genetically modified (GM) organism.

(a) T2A-1 rice is described as insect-resistant as it produces a chemical that is toxic to insect pests.

Outline how genetic engineering gave T2A-1 rice the trait of insect resistance.

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(b) Countries vary in the extent to which they grow GM varieties instead of traditional crop varieties.

- The USA, Canada and countries in South America plant GM varieties on a large proportion of their crop-growing land, reaching 94% for soybeans in the USA.
- In most European countries, 0% of the crop-growing land is planted with GM varieties.

Scientists used computer models to predict the effects of a global ban that reduced the cultivation of GM crops to 0% everywhere.

Table 4.1 shows the results of this modelling on world crop yields and the purchase price of each crop.

Table 4.1

crop	percentage change in crop yield	percentage change in purchase price
rapeseed	−0.14	+1.96
rice	−0.12	+1.58
soybeans	−1.40	+4.05

- (i) Explain what the data in Table 4.1 show about the **social** implications of growing GM crops.

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- (ii) A worldwide ban on growing GM crop varieties would mean that more land would be needed to grow traditional crops, such as Shanyou 2. This would involve converting forest and grassland to crop-growing land.

Table 4.2 shows the predicted changes in carbon dioxide emissions associated with this change in use of land.

Table 4.2

change in use of land	change in carbon dioxide emissions /million kg CO ₂
forest to crop-growing land	+ 608 726
grassland to crop-growing land	+ 276 042

Discuss what the data in Table 4.2 indicate about the environmental implications of growing GM crops.

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[Total: 8]